



UNIVERSIDAD DE ANTIOQUIA
1803

HEP computer tools

from SARAH and beyond

Diego Restrepo
2nd CONHEP school

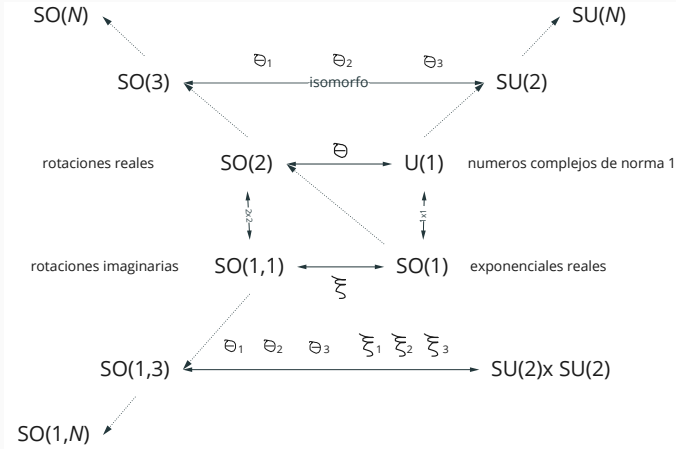
Instituto de Física
Universidad de Antioquia
Phenomenology Group
<http://gfif.udea.edu.co>

<https://ibug.io/blog/2019/05/mathematica-keygen/>

[https://docs.google.com/presentation/d/
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<https://clustercien.udea.edu.co/jupyter>

$$L = \frac{1}{2} m \mathbf{v} \cdot \mathbf{v} - V(|\mathbf{r}|).$$



$$U = \exp \left(i \sum_j T_j \theta^j \right), \quad (1)$$

where θ^j are the parameters of the transformation and T_j are the generators.

Consider the 1×1

$$K = -i, \quad (2)$$

which generates an element of dilaton group, $SO(1)$, $R(\xi)$

$$\lambda(\xi) = e^{\xi}, \quad (3)$$

which are just the group of the real exponentials. Such a number can be transformed as

$$x \rightarrow x' = e^{\xi} x, \quad (4)$$

that corresponds to a boost by e^{ξ} . We can define the invariant scalar product just as the division of real numbers, such that

$$x \cdot y \rightarrow x' \cdot y' \equiv \frac{x'}{y'} = \frac{e^{\xi} x}{e^{\xi} y} = \frac{x}{y} = x \cdot y. \quad (5)$$

We want to obtain a 2×2 representation of the algebra

$$K = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \rightarrow K^2 = -\mathbf{1}, \quad (6)$$

which can generate an element of the group SO(1, 1) with *parameter* ξ

$$\Lambda = \exp(i\xi K) = \begin{pmatrix} \cosh \xi & \sinh \xi \\ \sinh \xi & \cosh \xi \end{pmatrix}, \quad (7)$$

The transformation of a timelike and a spacelike coordinates, are ($c = 1$)

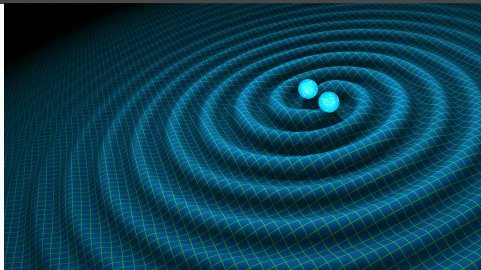
$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} x^0 \\ x^1 \end{pmatrix} \rightarrow \begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix} = \begin{pmatrix} \cosh \xi & \sinh \xi \\ \sinh \xi & \cosh \xi \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix}$$
$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad \mu = 0, 1.$$

$$\cosh \xi = \gamma = \frac{1}{\sqrt{1 - v^2}}$$

Special: parameter ξ or v is constant, e.g, inertial system invariance: *Global* conservation of E and $\mathbf{p}$ (still action at a distance!)

General: parameter $\xi(t, \mathbf{x})$ or $v(t, \mathbf{x})$ is local, e.g, accelerated system invariance: *Local* conservation of E and $\mathbf{p}$

Inestability of binary particle systems



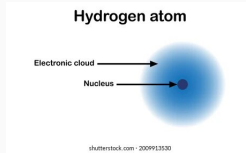
Gravitational wave discovery by LIGO



credits: science.org

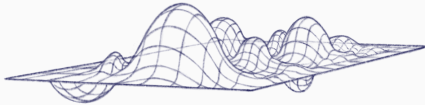
Noether's paradigm → Lagrangian formulation of classical field theory

U(1): From special θ to general $\theta(t, \mathbf{x})$



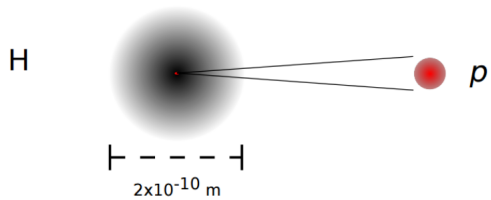
What is a *particle wavicle*? <https://www.quantamagazine.org/what-is-a-particle-20201112/>

Is a “Quantum Excitation of a Field”



Is a “Irreducible Representation of a Group”



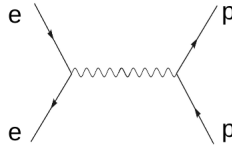
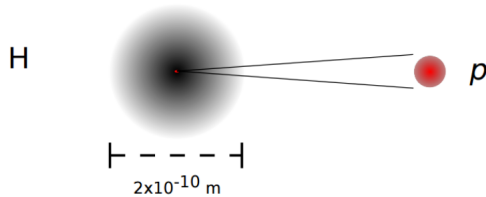


Interacción $\rightarrow$ Fuerza = $\Delta p / \Delta t$

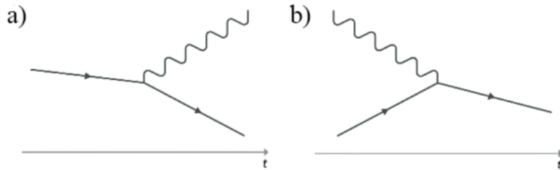
Introducción

Campos de materia $\longrightarrow$

Campos de radiación $\sim\sim\sim$



[doi:10.1088/1742-6596/1287/1/012045](https://doi.org/10.1088/1742-6596/1287/1/012045)



Emisión

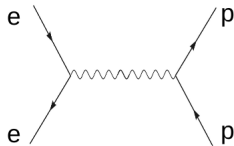
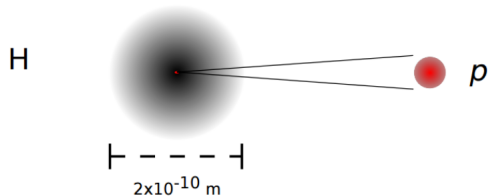
absorción

Interacción $\rightarrow$ Fuerza = $\Delta p / \Delta t$

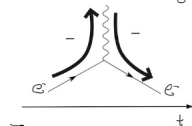
Introducción

Campos de materia $\longrightarrow$

Campos de radiación $\sim\sim\sim$



Single charge



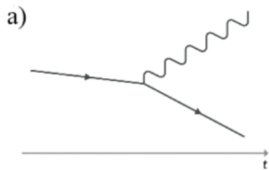
Fuerza $\frac{\Delta p}{\Delta t} \neq 0$

$$(e^-)^* = e^+$$

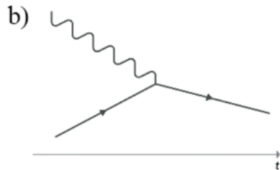
$\sim\sim\sim$ fotón neutro



[doi:10.1088/1742-6596/1287/1/012045](https://doi.org/10.1088/1742-6596/1287/1/012045)



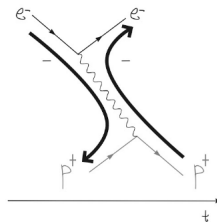
Emisión



absorción

$$e^- \rightarrow e^{-iEt}$$

$$e^+ \rightarrow e^{-iE(-t)}$$



Under a general Lorentz transformation we have for a **pure upperscript** 4-vector

$$A^\mu(x) \rightarrow A'^\mu(x) = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x), \quad (8)$$

where $\mu = 0, 1, 2, 3$. A **pure underscript** 4-vector is

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = (\partial_0, \nabla). \quad (9)$$

From $x'^\mu = \Lambda^\mu{}_\nu x^\nu$

$$\frac{1}{x'^\mu} = (\Lambda^{-1})^\nu{}_\mu \frac{1}{x^\nu}, \quad (10)$$

the tranformation properties for a $\partial_\mu = \partial/\partial x^\mu$, are

$$\partial'_\mu = \partial_\nu (\Lambda^{-1})^\nu{}_\mu. \quad (11)$$

In this way, the invariant scalar product between the 4-vector field and the four-gradient is just

$$\partial_\mu A^\mu \rightarrow \partial'_\mu A'^\mu = \partial_\mu A^\mu. \quad (12)$$

Photon: Representation of the Poincaré Group which transform as a vector under $SO(1,3)$

Name	Symbol	$SO(1,3)$
Photon	A^μ	$\Lambda^\mu{}_\nu A^\nu$
4-gradient	∂_μ	$\partial_\nu (\Lambda^{-1})^\nu{}_\mu$

Table 1: Scalar products: $\partial_\mu A^\mu$, $A^\nu A_\nu$, $\partial_\mu \partial^\mu$

Name	Symbol	$SU(N)$
scalar N -plet	ψ	$U\psi$
scalar anti- N -plet	$\psi^\dagger$	$\psi^\dagger U^\dagger$

Table 2: Scalar products: $\psi^\dagger \psi$

Photon: $\hat{p} \oplus \text{SO}(1,3) = i\partial^\mu \oplus \text{SO}(1,3) \rightarrow iD^\mu \oplus \text{SO}(1,3)$

Name	Symbol	SO(1,3)
Photon	A^μ	$\Lambda^\mu{}_\nu A^\nu$
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Table 2: Scalar products: $\psi^\dagger \psi$

Name	Symbol	SL(2, C)	$U(1)_Q$
e_L : electron left	ξ_α	$S_\alpha^\beta \xi_\beta$	$e^{i\theta} \xi_\alpha$
$(e_L)^\dagger$: positron right	$(\xi_\alpha)^\dagger = \xi_{\dot{\alpha}}^\dagger$	$\xi_{\dot{\beta}}^\dagger [S^\dagger]_{\dot{\alpha}}^{\dot{\beta}}$	$\xi_{\dot{\alpha}}^\dagger e^{-i\theta}$
e_R : electron right	$(\eta^\alpha)^\dagger = \eta^{\dagger\dot{\alpha}}$	$[(S^{-1})^\dagger]_{\dot{\beta}}^{\dot{\alpha}} \eta^{\dagger\dot{\beta}}$	$e^{i\theta} \eta^{\dagger\dot{\alpha}}$
$(e_R)^\dagger$: positron left	η^α	$\eta^\beta [S^{-1}]_\beta^\alpha$	$e^{-i\theta} \eta^\alpha$

Table 3: electron **left**: $SL(2, C) \times U(1)_Q$ (subscript) and positron **left**: $SL(2, C) \times U(1)_Q$ (superscript)

Scalar products

- ~~$U(1)$~~ Majorana scalars: $\xi^\alpha \xi_\alpha + \xi_{\dot{\alpha}}^\dagger \xi^{\dagger\dot{\alpha}}, \eta^\alpha \eta_\alpha + \eta_{\dot{\alpha}}^\dagger \eta^{\dagger\dot{\alpha}}.$
- Dirac scalar: $\eta^\alpha \xi_\alpha + \xi_{\dot{\alpha}}^\dagger \eta^{\dagger\dot{\alpha}}.$
- Tensor under subgroup $SL(2, C)$ but vector under $SO(1, 3)$: $S^{\dagger\dot{\alpha}}_{\dot{\beta}} \bar{\sigma}^\mu{}^{\dot{\beta}\beta} S_{\beta}^\alpha = \Lambda^\mu{}_\nu \bar{\sigma}^\nu{}^{\dot{\alpha}\alpha}$

$$\sigma^0 = \mathbb{1},$$

$$\bar{\sigma}^\mu = (\sigma^0, -\boldsymbol{\sigma}),$$

$$\sigma^\mu = (\sigma^0, \boldsymbol{\sigma}).$$

Name	Symbol	SL(2, C)	U(1) _Q
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Table 4: electron **left**: $SL(2, C) \times U(1)_Q$ (subscript) and positron **left**: $SL(2, C) \times U(1)_Q$ (superscript)

General theory: QED $\rightarrow D_\mu = i\partial_\mu - ieA_\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$$\xi_\alpha \rightarrow \xi'_\alpha = e^{i\theta(x)} \xi_\alpha$$

$$\eta^\alpha \rightarrow \eta'^\alpha = e^{-i\theta(x)} \eta^\alpha$$

$$D_\mu \xi_\alpha \rightarrow (D_\mu \xi_\alpha)' = e^{i\theta(x)} D_\mu \xi_\alpha$$

$$D_\mu \eta^\alpha \rightarrow (D_\mu \eta^\alpha)' = e^{-i\theta(x)} D_\mu \eta^\alpha$$

$$\mathcal{L} = i \xi_{\dot{\alpha}}^\dagger \bar{\sigma}^\mu{}^{\dot{\alpha}\alpha} D_\mu \xi_\alpha + i \eta^{\dagger\dot{\alpha}} \sigma^\mu{}_{\dot{\alpha}\alpha} D_\mu \eta^{\dagger\dot{\alpha}} - m \left[\eta^\alpha \xi_\alpha + \xi_{\dot{\alpha}}^\dagger \eta^{\dagger\dot{\alpha}} \right] - \frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

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Dirac spinor

$$\begin{aligned}
 \xi_\alpha &\rightarrow \xi'_\alpha = e^{i\theta(x)} \xi_\alpha & \eta^\alpha &\rightarrow \eta'^\alpha = e^{-i\theta(x)} \eta^\alpha \\
 D_\mu \xi_\alpha &\rightarrow (D_\mu \xi_\alpha)' = e^{i\theta(x)} D_\mu \xi_\alpha & D_\mu \eta^\alpha &\rightarrow (D_\mu \eta^\alpha)' = e^{-i\theta(x)} D_\mu \eta^\alpha \\
 \mathcal{L} &= i\bar{\xi}^\dagger \bar{\sigma}^\mu D_\mu \xi + i\eta^\dagger \sigma^\mu D_\mu \eta - m \left[\eta^\dagger \xi + \xi^\dagger \eta \right] - \frac{1}{4} F^{\mu\nu} F_{\mu\nu} \\
 \mathcal{L} &= i\bar{\psi} \gamma^\mu D_\mu \psi - m\bar{\psi} \psi - \frac{1}{4} F^{\mu\nu} F_{\mu\nu} .
 \end{aligned}$$

$$\begin{aligned}
 \psi &= \begin{pmatrix} e_L \\ e_R \end{pmatrix} = \begin{pmatrix} \xi \\ \eta^\dagger \end{pmatrix} \\
 \gamma^\mu &= \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \\
 \bar{\psi} &= \psi^\dagger \gamma^0 .
 \end{aligned}$$

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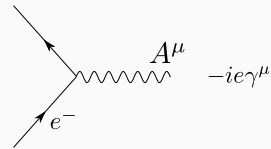
$$D_\mu \xi_\alpha \rightarrow (D_\mu \xi_\alpha)' = e^{i\theta(x)} D_\mu \xi_\alpha$$

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$$\mathcal{L} = i(e_L)^\dagger \bar{\sigma} \cdot D_\mu e_L + i(e_R)^\dagger \sigma^\mu D_\mu e_R - m \left[(e_R)^\dagger e_L + (e_L)^\dagger e_R \right] - \frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

$$\mathcal{L} = i\bar{\psi} \gamma^\mu D_\mu \psi - m\bar{\psi} \psi - \frac{1}{4} F^{\mu\nu} F_{\mu\nu} \rightarrow e\bar{\psi} \gamma^\mu \psi A_\mu$$

Dirac spinor



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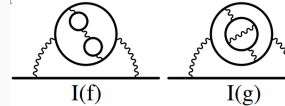
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$$\mathcal{L} = i\bar{\psi} \gamma^\mu D_\mu \psi - m\bar{\psi} \psi - \frac{1}{4} F^{\mu\nu} F_{\mu\nu} \rightarrow e\bar{\psi} \gamma^\mu \psi A_\mu$$

Dirac spinor

non-bare

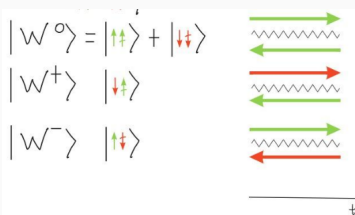


$SU(2)_L \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i$: 17 years later... (stages of grief < 1967)

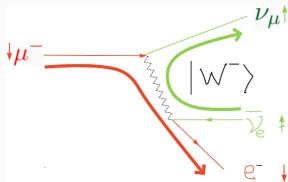
Not mass,

Field	Lorentz	$SU(2)_L$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2

Denial



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu}$$

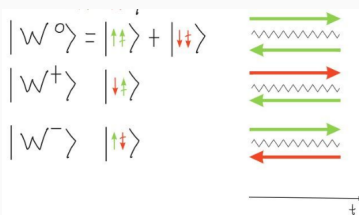


$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 17 years later... (stages of grief < 1967)

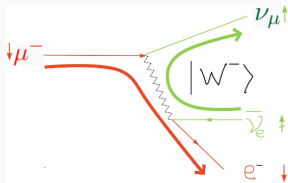
Not mass, **hypercharge**,

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$

Denial



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}$$

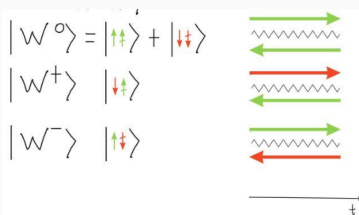


$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 17 years later... (stages of grief < 1967)

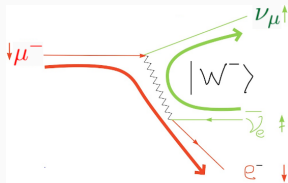
Not mass, hypercharge, not Dirac

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1

Denial



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R$$

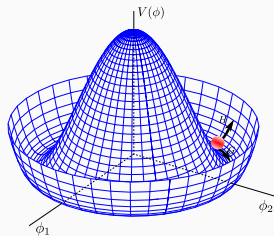


$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 17 years later... (stages of grief > 1967)

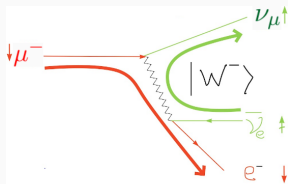
Higgs mechanism: tachyonic mass $\mu^2 < 0$, and condensate

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \left[\frac{H(x) + v}{\sqrt{2}} \right] \exp \left[i \frac{\tau^i}{2} G_i(x) \right]$	-	2	$1/2$

Contempt



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R + h(e_R)^\dagger \Phi^\dagger L - (D^\mu \Phi)^\dagger D_\mu \Phi - \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2$$



$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 17 years later... (stages of grief $\rightarrow$ 1967)

Higgs mechanism: tachyonic mass $\mu^2 < 0$, and condensate

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \left[\frac{H(x) + v}{\sqrt{2}} \right] \exp \left[i \frac{\tau^i}{2} G_i(x) \right]$	-	2	$1/2$

Contempt

$$\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix},$$

$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R + h(e_R)^\dagger \Phi^\dagger L - (D^\mu \Phi)^\dagger D_\mu \Phi - \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2$$

$$\Phi \rightarrow \Phi' = \exp \left[i \frac{\tau^i}{2} \theta_i(x) \right] \Phi = \frac{1}{\sqrt{2}} [H(x) + v]$$

$$\mathcal{L} = \bar{\psi} \gamma^\mu \partial_\mu \psi - m_e \bar{\psi} \psi - i(\nu_L)^\dagger \bar{\sigma}^\mu \partial_\mu \nu_L + \frac{1}{2} \partial^\mu H \partial_\mu H + \frac{e}{\cos \theta_W \sin \theta_W} \bar{\nu}_L \nu_L Z_\mu + \dots$$

$$- \frac{1}{2} m_H^2 H^2 \left(1 + \frac{H}{v} + \frac{H^2}{4v^2} \right) + \left(m_W^2 W^{\mu-} W_\mu^+ + \frac{1}{2} m_Z^2 Z^\mu Z_\mu \right) \left(1 + 2 \frac{H}{v} + \frac{H^2}{v^2} \right) + \frac{m_e}{v} \bar{\psi} \psi H$$

$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 21 years later... (stages of grief $\rightarrow$ 1971)

Z and W phenomenology and discovery

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \left[\frac{H(x) + v}{\sqrt{2}} \right] \exp \left[i \frac{\tau_i}{2} G_i(x) \right]$	-	2	$1/2$

Bargaining



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R + h(e_R)^\dagger \Phi^\dagger L - (D^\mu \Phi)^\dagger D_\mu \Phi - \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2$$

$$\Phi \rightarrow \Phi' = \exp \left[i \frac{\tau_i}{2} \theta_i(x) \right] \Phi = \frac{1}{\sqrt{2}} [H(x) + v]$$

$$\mathcal{L} = i\bar{\psi}\gamma^\mu \partial_\mu \psi - m_e \bar{\psi}\psi - i(\nu_L)^\dagger \bar{\sigma}^\mu \partial_\mu \nu_L + \frac{1}{2} \partial^\mu H \partial_\mu H + \frac{e}{\cos \theta_W \sin \theta_W} \bar{\nu}_L \nu_L Z_\mu + \dots$$

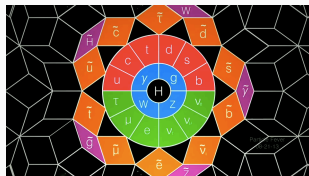
$$- \frac{1}{2} m_H^2 H^2 \left(1 + \frac{H}{v} + \frac{H^2}{4v^2} \right) + \left(m_W^2 W^{\mu-} W_\mu^+ + \frac{1}{2} m_Z^2 Z^\mu Z_\mu \right) \left(1 + 2 \frac{H}{v} + \frac{H^2}{v^2} \right) + \frac{m_e}{v} \bar{\psi}\psi H$$

$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 32 years later... (stages of grief $\rightarrow$ 1982)

Hierarchy problem

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \left[\frac{H(x) + v}{\sqrt{2}} \right] \exp \left[i \frac{\tau^i}{2} G_i(x) \right]$	-	2	$1/2$

Depression



credit: quantumdiaries.org

$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R + h(e_R)^\dagger \Phi^\dagger L - (D^\mu \Phi)^\dagger D_\mu \Phi - \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2$$

$$\Phi \rightarrow \Phi' = \exp \left[i \frac{\tau^i}{2} \theta_i(x) \right] \Phi = \frac{1}{\sqrt{2}} [H(x) + v]$$

$$\mathcal{L} = i\bar{\psi}\gamma^\mu \partial_\mu \psi - m_e \bar{\psi}\psi - i(\nu_L)^\dagger \bar{\sigma}^\mu \partial_\mu \nu_L + \frac{1}{2} \partial^\mu H \partial_\mu H + \frac{e}{\cos \theta_W \sin \theta_W} \bar{\nu}_L \nu_L Z_\mu + \dots$$

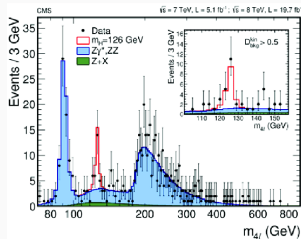
$$- \frac{1}{2} m_H^2 H^2 \left(1 + \frac{H}{v} + \frac{H^2}{4v^2} \right) + \left(m_W^2 W^{\mu-} W_\mu^+ + \frac{1}{2} m_Z^2 Z^\mu Z_\mu \right) \left(1 + 2 \frac{H}{v} + \frac{H^2}{v^2} \right) + \frac{m_e}{v} \bar{\psi}\psi H$$

$SU(2)_L \times U(1)_Y \rightarrow D_\mu = \mathbf{1}\partial_\mu - ig_2 \frac{\tau_i}{2} W_\mu^i - ig_1 B_\mu$: 62 years later... (stages of grief $\rightarrow$ 2012)

Higgs discovery!

Field	Lorentz	$SU(2)_L$	$U(1)_Y$
$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	ξ_α	2	$-1/2$
$(e_R)^\dagger$	η^α	1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \left[\frac{H(x) + v}{\sqrt{2}} \right] \exp \left[i \frac{\tau_i}{2} G_i(x) \right]$	-	2	$1/2$

Acceptance



$$\mathcal{L} = i(L)^\dagger \bar{\sigma}^\mu D_\mu L - \frac{1}{4} W_{\mu\nu}^i W_i^{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - i(e_R)^\dagger \sigma^\mu D_\mu e_R + \boxed{h(e_R)^\dagger \Phi^\dagger L} + (D^\mu \Phi)^\dagger D_\mu \Phi - \boxed{\mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2}$$

$$\Phi \rightarrow \Phi' = \exp \left[i \frac{\tau_i}{2} \theta_i(x) \right] \Phi = \frac{1}{\sqrt{2}} [H(x) + v]$$

$$\mathcal{L} = i\bar{\psi}\gamma^\mu \partial_\mu \psi - m_e \bar{\psi}\psi - i(\nu_L)^\dagger \bar{\sigma}^\mu \partial_\mu \nu_L + \frac{1}{2} \partial^\mu H \partial_\mu H + \frac{e}{\cos \theta_W \sin \theta_W} \bar{\nu}_L \nu_L Z_\mu + \dots$$

$$- \frac{1}{2} m_H^2 H^2 \left(1 + \frac{H}{v} + \frac{H^2}{4v^2} \right) + \left(m_W^2 W^{\mu-} W_\mu^+ + \frac{1}{2} m_Z^2 Z^\mu Z_\mu \right) \left(1 + 2 \frac{H}{v} + \frac{H^2}{v^2} \right) + \frac{m_e}{v} \bar{\psi}\psi H$$

First generation of Standard Model (SM)

Field	Lorentz	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$
Q	ξ_α^1	3	2	$1/6$
L	ξ_α^2	1	2	$-1/2$
$(u_R^-)^\dagger$	η_1^α	$\bar{\mathbf{3}}$	1	$-2/3$
$(d_R^-)^\dagger$	η_2^α	$\bar{\mathbf{3}}$	1	$1/3$
$(e_R^-)^\dagger$	η_3^α	1	1	1
Φ	-	1	2	$1/2$

Table 5: Fundamental Fields of SM

$$\tilde{\Phi} = i\sigma_2 \Phi^* = \begin{pmatrix} \phi^{0*} & -\phi^- \end{pmatrix}^T \quad L = \begin{pmatrix} \nu_L & e_L \end{pmatrix}^T \quad (13)$$

$$(e_R)^\dagger \Phi^\dagger L = (e_R)^\dagger \Phi^\dagger L = (e_R)^\dagger \epsilon_{ab} \tilde{\Phi}^a L^b, \quad (14)$$

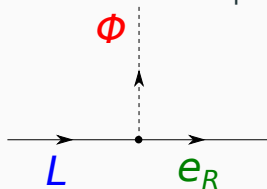
$$-Y_R - Y_\Phi + Y_L = 0$$

$$\tilde{\Phi} = i\sigma_2 \Phi^* = \begin{pmatrix} \phi^{0*} & -\phi^- \end{pmatrix}^T$$

$$L = \begin{pmatrix} \nu_L & e_L \end{pmatrix}^T \quad (13)$$

$$(e_R)^\dagger \tilde{\Phi} \cdot L, \quad (14)$$

Which can be represented as the Kirchoff's law:



$$\begin{aligned} -Y_R - Y_\Phi + Y_L &= 0 \\ Y_L &= Y_\Phi + Y_R \end{aligned}$$

Install compilers:

```
sudo apt install build-essential gfortran feynmf
```

Download all the HEP-tools in one step:

```
git clone --recursive https://github.com/restrepo/BSM-Submodules.git  
cd BSM-Submodules/  
emacs SARAH/Models/SSDM/SSDM.m
```

$$SU(3)_c \times SU(2)_L \times U(1)_Y \times Z_2$$

$$\begin{aligned} \mathcal{D}_\mu = & \partial_\mu - ig_1 Y B_\mu \\ & - ig_2 T W_\mu^B \\ & - ig_3 \Lambda G_\mu. \end{aligned}$$

$$B \rightarrow B_\mu, \tilde{B} \quad \Longrightarrow \quad \text{VB, FB}$$

```
Off[General::spell]

Model`Name = "SSDM";
Model`NameLaTeX = "Singlet scalar Dark Matter";
Model`Authors = "Diego Restrepo ...";
Model`Date = "2015-11-16";

(* 2013-01-24: ... )

(* Global Symmetries *)

Global[[1]] = {Z[2], Z2};

(* Gauge Groups *)

Gauge[[1]]={B, U[1], hypercharge, g1,False,1};
Gauge[[2]]={WB, SU[2], left, g2,True,1};
Gauge[[3]]={G, SU[3], color, g3,False,1};
```

	N_F	Lorentz	Y	$SU(2)_L$	$SU(3)_c$	Z_2
Q	3	$\xi_{1\alpha} : (u_L d_L)^T$	1/6	2	3	+
L	3	$\xi_{2\alpha} : (\nu_L e_L)^T$	-1/2	2	1	+
d^c	3	$\eta_1^\alpha : (d_R)^\dagger$	1/3	1	$\bar{3}$	+
u^c	3	$\eta_2^\alpha : (u_R)^\dagger$	-2/3	1	$\bar{3}$	+
e^c	3	$\eta_3^\alpha (e_R)^\dagger$	1	1	1	+
H	1	$(H^+ H^0)$	1/2	2	1	+
S	1	s	0	1	1	-

$$\text{conj}[H] = \tilde{H} = \begin{pmatrix} H^{0*} \\ -H^- \end{pmatrix}$$

$$\mathcal{L} = (\mathcal{L}_C + \text{h.c}) \\ + \mathcal{L}_R,$$

$$\mathcal{L}_C = -Y_e e^c \tilde{H} \cdot L - Y_d d^c \tilde{H} \cdot Q - Y_u u^c H \cdot Q,$$

$$\mathcal{L}_R = -\mu^2 \tilde{H} \cdot H - \lambda_1 (\tilde{H} \cdot H)^2 \\ - M_S^2 S^2 - \lambda_{SH} S^2 \tilde{H} \cdot H - \lambda_S S^4.$$

$$Y_e \rightarrow N_F \times N_F, \dots$$

```
(* Matter Fields *)

FermionFields[[1]] = {q, 3, {uL, dL},      1/6, 2,  3,1};
FermionFields[[2]] = {l, 3, {vL, eL},      -1/2, 2,  1,1};
FermionFields[[3]] = {dc, 3, conj[dR],      1/3, 1, -3,1};
FermionFields[[4]] = {uc, 3, conj[uR],      -2/3, 1, -3,1};
FermionFields[[5]] = {ec, 3, conj[eR],      1, 1,  1,1};

ScalarFields[[1]] = {H, 1, {Hp, H0},      1/2, 2,  1,1};
ScalarFields[[2]] = {S, 1, ss,            0, 1,  1,-1};
RealScalars = {S};

(*-----*)
(*  DEFINITION  *)
(*-----*)

NameOfStates={GaugeES, EWSB};

(* ----- Before EWSB ----- *)

DEFINITION[GaugeES][LagrangianInput]= {
  {LagHC, {AddHC->True}},
  {LagNoHC,{AddHC->False}}
};

LagHC =  -(Ye ec.conj[H].l + Yd dc.conj[H].q + Yu uc.H.q);

LagNoHC = -(mu2 conj[H].H + Lambda1/2 conj[H].H.conj[H].H
           + MS2/2 S.S + LamSH S.S.conj[H].H + LamS/2 S.S.S.S);
```

$$\begin{pmatrix} B^\mu \\ W_3^\mu \end{pmatrix} = Z^\gamma Z \begin{pmatrix} A^\mu \\ Z^\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} A^\mu \\ Z^\mu \end{pmatrix}$$

$$H^0 = \frac{iA + (h + v)}{\sqrt{2}}.$$

$$d_L = V_d D_L, \quad (d_R)^\dagger = D_R^c U_d,$$

$$V_d \rightarrow N_F \times N_F, \dots$$

$$\psi_d = \begin{pmatrix} \xi_{1\alpha} \\ (\eta_1^\alpha)^\dagger \end{pmatrix} = \begin{pmatrix} D_L \\ D_R \end{pmatrix}$$

(* Gauge Sector *)

```
DEFINITION[EWSB][GaugeSector] =
{
  {{VB,VWB[3]},{VP,VZ},{ZZ},
   {{VWB[1],VWB[2]},{VWp,conj[VWp]},{ZW}
};
```

(* ----- VEVs ----- *)

```
DEFINITION[EWSB][VEVs]=
{{H0,{v,1/Sqrt[2]},{Ah,[ImaginaryI]/Sqrt[2]},{hh,1/Sqrt[2]}}};
```

```
DEFINITION[EWSB][MatterSector]=
{{{dL},{conj[dR]}},{DL,Vd},{DRc,Ud}},
{{{uL},{conj[uR]}},{UL,Vu},{URc,Uu}},
{{{eL},{conj[eR]}},{EL,Ve},{ERc,Ue}}};
```

(*-----*)

(* Dirac-Spinors *)

(*-----*)

```
DEFINITION[EWSB][DiracSpinors]={
  Fd ->{ DL, conj[DRc]},
  Fe ->{ EL, conj[ERc]},
  Fu ->{ UL, conj[URc]},
  Fv ->{ vL, 0}};
```

$$\begin{pmatrix} B^\mu \\ W_3^\mu \end{pmatrix} = Z^\gamma Z \begin{pmatrix} A^\mu \\ Z^\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} A^\mu \\ Z^\mu \end{pmatrix}$$

$$H^0 = \frac{iA + (h + v)}{\sqrt{2}}.$$

$$d_L = V_d D_L,$$

$$(d_R)^\dagger = D_R^c U_d,$$

$$V_d \rightarrow N_F \times N_F, \dots$$

Chuck Norris fact of the day

*Chuck Norris lost his virginity
before his dad*



From A. Vicente

(* Gauge Sector *)

DEFINITION[EWSB][GaugeSector] =

```
{ {VB, VWB[3]}, {VP, VZ}, ZZ },
{ {VWB[1], VWB[2]}, {VWp, conj[VWp]}, ZW }
};
```

(* ----- VEVs ----- *)

DEFINITION[EWSB][VEVs]=

```
{ {h0, {v, 1/Sqrt[2]}}, {Ah, \[ImaginaryI]/Sqrt[2]}, {hh, 1/Sqrt[2]} };
```

DEFINITION[EWSB][MatterSector]=

```
{ { {dL}, {conj[dR]} }, { {DL, Vd}, {DRc, Ud} },
{ {uL}, {conj[uR]} }, { {UL, Vu}, {URc, Uu} },
{ {eL}, {conj[eR]} }, { {EL, Ve}, {ERc, Ue} } };
```

(*-----*)

(* Dirac-Spinors *)

(*-----*)

DEFINITION[EWSB][DiracSpinors]={

```
Fd ->{ DL, conj[DRc]},
Fe ->{ EL, conj[ERc]},
Fu ->{ UL, conj[URc]},
Fv ->{ vL, 0}};
```

```
...
{g1,      { Description -> "Hypercharge-Coupling"}},
{g2,      { Description -> "Left-Coupling"}},
{g3,      { Description -> "Strong-Coupling"}},
...
{v,       { Description -> "EW-VEV",
           DependenceNum -> Sqrt[4*Mass[VWp]^2/(g2^2)],
           DependenceSPheno -> None }},

{ThetaW,  { Description -> "Weinberg-Angle",
           DependenceNum -> ArcSin[Sqrt[1 - Mass[VWp]^2/Mass[VZ]^2]]}},

{ZZ, {Description -> "Photon-Z Mixing Matrix"}},

...
```

```
...  
{Description -> "Photon-Z Mixing Matrix",  
  Dependence -> {{Cos[ThetaW], -Sin[ThetaW]},  
                 {Sin[ThetaW], Cos[ThetaW]}},  
  Real -> True,  
  LaTeX -> "Z^{\gamma Z}",  
  LesHouches -> None,  
  OutputName -> ZZ }},  
...
```

```
...
ParticleDefinitions[EWSB] = {

  {hh    , { Description -> "Higgs",
             PDG -> {25},
             PDG.IX -> {101000001},
             Mass -> LesHouches,
             FeynArtsNr -> 1,
             LaTeX -> "h",
             ElectricCharge -> 0,
             LHPC -> {1},
             OutputName -> "h"  }},

  {ss    , { Description -> "Singlet",
             PDG -> {6666635},
             PDG.IX -> {101000002},
             FeynArtsNr -> 10,
             Mass -> LesHouches,
             LaTeX -> "S",
             ElectricCharge -> 0,
             LHPC -> {"gold"},
             OutputName -> "Ss" }},

  ...
}
...
```

```
OnlyLowEnergySPheno = True;  
  
MINPAR={{1,Lambda1IN},  
        {2,LamSHIN},  
        {3,LamSIN},  
        {4,MSinput}  
        };  
  
...  
ListDecayParticles = {Fu,Fe,Fd,hh};  
ListDecayParticles3B = {{Fu,"Fu.f90"},{Fe,"Fe.f90"},{Fd,"Fd.f90"}};  
  
....  
  
DefaultInputValues = {Lambda1IN -> 0.28, LamSHIN -> 0.01, LamSIN -> 0,  
                       MSinput -> 200};
```

Implicit files without editors

```
cat << EOF > kk.txt  
Hello world  
EOF
```

Listing 1: Creates kk.txt file

```
math << EOF  
2+2  
EOF
```

Listing 2: commands expecting input files

```
Mathematica 11.0.0 for Linux x86 (64-bit)  
Copyright 1988-2016 Wolfram Research, Inc.  
  
In[1]:=   
Out[1]= 4  
  
In[2]:=
```

Verifying Index Contractions

$$u^c \cdot H \cdot q = \delta_\alpha^\beta \epsilon_{ab} u_\alpha^c H^a q^{\beta b} \\ = \bar{3} \times 3 \otimes 2 \times 2.$$

```
math<<EOF
<<./SARAH/SARAH.m
Start["SSDM"]
MakeIndexStructure[{u, H, q}] (* uc.H.q *)
EOF
...
Out[3]: Delta[col1, col3] epsTensor[lef2, lef3]
```

Explicit index contraction in SSDM.m

```
Delta[col1, col3] epsTensor[lef2, lef3] uc.H.q
```

See https://gitlab.in2p3.fr/goodsell/sarah/wikis/Automatic_index_contraction

Check SARAH

```
math << EOF
<<./SARAH/SARAH.m
Start["SSDM"]
MakeSPheno[]
EOF
```

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...

In[1]:= SARAH 4.14.1

by Florian Staub, 2018

...

In[2]:= Preparing arrays

...

Model files loaded

Model : SSDM

Author(s): Diego Restrepo (based on SM model by F.Staub)

Date : 2015-11-16

Loading Susyno functions for the handling of Lie Groups

Based on Susyno v.2.0 by Renato Fonseca (1106.5016)

webpage: web.ist.utl.pt/renato.fonseca/susyno.html

...

Finished! SPheno code generated in 170.872s

....

The following steps are now necessary to implement the model in SPheno:

Check SPheno

```
cp -r SARAH/Output/SSDM/EWSB/SPheno SPheno/SSDM
cd SPheno
make Model=SSDM # Be sure that Makefile use gfortran!
```

```
cd SSDM ; make F90=gfortran version=400.00
make[1]: Entering directory '****/BSM-Submodules/SPHENO/SSDM'
.
.
.
make[2]: Leaving directory '****/BSM-Submodules/SPHENO/SSDM'
gfortran -o SPhenoSSDM -g SPhenoSSDM.o ../lib/libSPhenoSSDM.a ../lib/libSPheno.a
mv SPhenoSSDM ../bin
rm SPhenoSSDM.o
make[1]: Leaving directory '****/BSM-Submodules/SPHENO/SSDM'
```

```
# Return to parent directory: BSM-Submodules
cd ../
```

Check micrOMEGAs

```
cd micromegas
make # Recompile everything!
make # twice
```

```
make -C CalcHEP_src MICROMEGAS=MICROMEGAS
...
#-----
# CalcHEP has compiled successfully and can be started.
# The manual can be found on the CalcHEP website:
#   http://theory.sinp.msu.ru/~pukhov/calchep.html
# The next step is typically to run
#   ./mkWORKdir <new_dir>
# where <new_dir> is the new directory where you will do
# your calculations. After creating this directory, you
# should cd into it and run calchep or calchep_batch.
# Please see the manual for further details.
#-----"
...
make[1]: Leaving directory '****/BSM-Submodules/micromegas/sources'
```

Build micrOMEGAs model

```
./newProject SSDM
cd .. # return to parent directory
```

Check micrOMEGAs II

```
math << EOF
<<./SARAH/SARAH.m
Start["SSDM"]
MakeCHep[]
EOF
```

Mathematica 11.0.0 for Linux x86 (64-bit)

...

Write main file for MicrOmegas

Done. Model files generated in 31.044s

Output is saved in ***/BSM-Submodules/SARAH/Output/SSDM/EWSB/CHep/

```
cp SARAH/Output/SSDM/EWSB/CHep/* micromegas/SSDM/work/models/
cd micromegas/SSDM/
cp work/models/*.cpp .
# check your micrOMEGAs version
make main=CalcOmega_with_DDetection_M0v5.cpp
```

make -C work

...

g++ -g -fPIC -o CalcOmega_with_DDetection_M0v5 CalcOmega_with_DDetection_M0v5.cpp ... -lpthread

cd ../../ #Return to parent directory

Check Madgraph

```
math << EOF
<<./SARAH/SARAH.m
Start["SSDM"]
MakeUFO[]
EOF
```

```
Mathematica 11.0.0 for Linux x86 (64-bit)
...
Writing effective diphoton and digluon vertices

Done. UFO files generated in 30.716s
Output is saved in ***/BSM-Submodules/SARAH/Output/SSDM/EWSB/UFO/
```

```
cp -r SARAH/Output/SSDM/EWSB/UFO/ madgraph/models/SSDM
madgraph/bin/mg5_aMC << EOF
import model SSDM
check u u~ > mu+ mu-
EOF
```

Process	Min element	Max element	Relative diff.	Result
u u~ > mu+ mu-	4.9949890843e-03	4.9949890843e-03	5.2093911919e-15	Passed
Summary: 1/1 passed, 0/1 failed				

```
cp SPheno/SSDM/Input_Files/LesHouches.in.SSDM .  
emacs LesHouches.in.SSDM
```



```

Block MODSEL      #
1 1               # 1/0: High/low scale input
2 1               # Boundary Condition
6 1               # Generation Mixing
Block SMINPUTS     # Standard Model inputs
2 1.166370E-05    # G_F,Fermi constant
3 1.187000E-01    # alpha_s(MZ) SM MSbar
4 9.118870E+01    # Z-boson pole mass
5 4.180000E+00    # m_b(mb) SM MSbar
6 1.735000E+02    # m_top(pole)
7 1.776690E+00    # m_tau(pole)
Block MINPAR       # Input parameters
1 2.8000000E-01   # Lambda1IN
2 1.0000000E-02   # LamSHIN
3 0.0000000E+00   # LamSIN
4 2.0000000E+02   # MSinput
Block SPhenoInput  # SPheno specific input
1 -1              # error level
2 0               # SPA conventions
7 0               # Skip 2-loop Higgs corrections
8 3               # Method used for two-loop calculation
9 1               # Gaugeless limit used at two-loop
10 0              # safe-mode used at two-loop
11 1              # calculate branching ratios
13 1              # 3-Body decays: none (0), fermion (1), scalar (2), both (3)
14 0              # Run couplings to scale of decaying particle
12 1.000E-04      # write only branching ratios larger than this value
15 1.000E-30      # write only decay if width larger than this value
16 1              # One-loop decays
18 2              # Matching order (0: automatic, 1: pole, 2: tree, 3: one-loop, 4: two-loop)
-:-- LesHouches.in.SSDM Top L16 (Fundamental)

```



```
Block MODSEL      #
1 1               # 1/0: High/low scale input
2 1               # Boundary Condition
6 1               # Generation Mixing
Block SMINPUTS    # Standard Model inputs
2 1.166370E-05    # G_F,Fermi constant
3 1.187000E-01    # alpha_s(MZ) SM MSbar
4 9.118870E+01    # Z-boson pole mass
5 4.180000E+00    # m_b(mb) SM MSbar
6 1.735000E+02    # m_top(pole)
7 1.776690E+00    # m_tau(pole)
Block MINPAR      # Input parameters
1 2.8000000E-01   # LambdaLIN
2 1.0000000E-02   # LamSHIN
3 0.0000000E+00   # LamSIN
4 2.0000000E+02   # MSinput
Block SPhenoInput # SPheno specific input
1 -1              # error level
2 0               # SPA conventions
7 0               # Skip 2-loop Higgs corrections
8 3               # Method used for two-loop calculation
9 1               # Gaugeless limit used at two-loop
10 0              # safe-mode used at two-loop
11 1              # calculate branching ratios
13 1              # 3-Body decays: none (0), fermion (1), scalar (2), both (3)
14 0              # Run couplings to scale of decaying particle
12 1.000E-04      # write only branching ratios larger than this value
15 1.000E-30      # write only decay if width larger than this value
16 1              # One-loop decays
10 2              # Matching order ( 2:automatic, 1:scale, 0:2, tree, and 5:two loop)
-:-- LesHouches.in.SSDM Top L16 (Fundamental)
```

Run SPheno

```
BSM-Submodules$ SPheno/bin/SPhenoSSDM LesHouches.in.SSDM
Calculating branching ratios and decay widths
Calculating one loop decays
Loop masses not calculated: tree-level masses used for kinematics
Loop masses not calculated: no U-factors are applied
Calculating one-loop decays of Fu
Calculating one-loop decays of Fe
Calculating one-loop decays of Fd
Calculating one-loop decays of hh
Calculating low energy constraints
Calculating unitarity constraints
Writing output files
Finished!
BSM-Submodules$ emacs SPheno.spc.SSDM
```



```
# SUSY Les Houches Accord 2 - SSDM Spectrum + Decays + Flavor Observables
# SPheno module generated by SARAH
# -----
# SPheno v4.0.3
# W. Porod, Comput. Phys. Commun. 153 (2003) 275-315, hep-ph/0301101
# W. Porod, F.Staub, Comput.Phys.Commun.183 (2012) 2458-2469, arXiv:1104.1573
# SARAH: 4.14.1
# F. Staub; arXiv:0806.0538 (online manual)
# F. Staub; Comput. Phys. Commun. 181 (2010) 1077-1086; arXiv:0909.2863
# F. Staub; Comput. Phys. Commun. 182 (2011) 808-833; arXiv:1002.0840
# F. Staub; Comput. Phys. Commun. 184 (2013) 1792-1809; arXiv:1207.0906
# F. Staub; Comput. Phys. Commun. 185 (2014) 1773-1790; arXiv:1309.7223
# Including the calculation of flavor observables based on the FlavorKit
# W. Porod, F. Staub, A. Vicente; Eur.Phys.J. C74 (2014) 8, 2992; arXiv:1405.1434
# Two-loop mass corrections to Higgs fields based on
# M. D. Goodsell, K. Nickel, F. Staub; Eur.Phys.J. C75 (2015) no.6, 290; arXiv:1411.0675
# M. D. Goodsell, K. Nickel, F. Staub; Eur.Phys.J. C75 (2015) no.1, 32; arXiv:1503.03098
# M. D. Goodsell, F. Staub; arXiv:1511.01904
#
# in case of problems send email to florian.staub@kit.edu and goodsell@lpthe.jussieu.fr
# -----
# Created: 19.09.2019, 22:47
Block SPINFO # Program information
1 SPhenoSARAH # spectrum calculator
2 v4.0.3 # version number of SPheno
9 4.14.1 # version number of SARAH
Block MODESEL # Input parameters
1 1 # GUT scale input
2 1 # Boundary conditions
6 1 # switching on flavour violation
Block MINPAR # Input parameters
1 2.80000000E-01 # Lambda1IN
2 1.00000000E-02 # LamSHIN
3 0.00000000E+00 # LamSIN
4 2.00000000E+02 # MSinput
Block gaugeGUT Q= -1.00000000E+00 # (GUT scale)
1 0.00000000E+00 # g1(Q)
2 0.00000000E+00 # g2(Q)
3 0.00000000E+00 # g3(Q)
Block SMINPUTS # SM parameters
-:--- SPheno.spc.SSDM Top L39 (Fundamental)
Beginning of buffer
```

```

3 3      9.95678124E-01 # Real(Yu(3,3),dp)
Block Yd Q= 1.60000000E+02 # (Renormalization Scale)
1 1      2.87184285E-05 # Real(Yd(1,1),dp)
1 2      0.00000000E+00 # Real(Yd(1,2),dp)
1 3      0.00000000E+00 # Real(Yd(1,3),dp)
2 1      0.00000000E+00 # Real(Yd(2,1),dp)
2 2      5.45650142E-04 # Real(Yd(2,2),dp)
2 3      0.00000000E+00 # Real(Yd(2,3),dp)
3 1      0.00000000E+00 # Real(Yd(3,1),dp)
3 2      0.00000000E+00 # Real(Yd(3,2),dp)
3 3      2.40086062E-02 # Real(Yd(3,3),dp)
Block Ye Q= 1.60000000E+02 # (Renormalization Scale)
1 1      2.93501725E-06 # Real(Ye(1,1),dp)
1 2      0.00000000E+00 # Real(Ye(1,2),dp)
1 3      0.00000000E+00 # Real(Ye(1,3),dp)
2 1      0.00000000E+00 # Real(Ye(2,1),dp)
2 2      6.06868478E-04 # Real(Ye(2,2),dp)
2 3      0.00000000E+00 # Real(Ye(2,3),dp)
3 1      0.00000000E+00 # Real(Ye(3,1),dp)
3 2      0.00000000E+00 # Real(Ye(3,2),dp)
3 3      1.02047490E-02 # Real(Ye(3,3),dp)

```

```

Block MASS # Mass spectrum
# PDG code mass particle
25 1.30287679E+02 # hh
6666635 2.83944658E+01 # ss
23 9.11887000E+01 # VZ
24 8.03497269E+01 # VWp
1 5.00000000E-03 # Fd_1
3 9.50000000E-02 # Fd_2
5 4.18000000E+00 # Fd_3
2 2.50000000E-03 # Fu_1
4 1.27000000E+00 # Fu_2
6 1.73500000E+02 # Fu_3
11 5.10998930E-04 # Fe_1
13 1.05658372E-01 # Fe_2
15 1.77669000E+00 # Fe_3

```

$$M_S = 28 \text{ GeV}$$

```

Block UDL MIX Q= 1.60000000E+02 # ( )
1 1      1.00000000E+00 # Real(ZDL(1,1),dp)
1 2      0.00000000E+00 # Real(ZDL(1,2),dp)
1 3      0.00000000E+00 # Real(ZDL(1,3),dp)
-:--- SPheno.spc.SSDM 9% L75 (Fundamental)

```

```

52 0.00000000E+00 # Ignore negative masses
53 0.00000000E+00 # Ignore negative masses at MZ
55 0.00000000E+00 # Calculate one loop masses
56 1.00000000E+00 # Calculate two-loop Higgs masses
57 1.00000000E+00 # Calculate low energy
60 1.00000000E+00 # Include kinetic mixing
65 1.00000000E+00 # Solution of tadpole equation
Block HiggsBoundsInputHiggsCouplingsFermions #
1.00000000E+00 0.00000000E+00 3 25 5 5 # h_1 b b coupling
1.00000000E+00 0.00000000E+00 3 25 3 3 # h_1 s s coupling
1.00000000E+00 0.00000000E+00 3 25 6 6 # h_1 t t coupling
1.00000000E+00 0.00000000E+00 3 25 4 4 # h_1 c c coupling
1.00000000E+00 0.00000000E+00 3 25 15 15 # h_1 tau tau coupling
1.00000000E+00 0.00000000E+00 3 25 13 13 # h_1 mu mu coupling
Block HiggsBoundsInputHiggsCouplingsBosons #
1.00000000E+00 3 25 24 24 # h_1 W W coupling
1.00000000E+00 3 25 23 23 # h_1 Z Z coupling
0.00000000E+00 3 25 23 22 # h_1 Z gamma coupling
1.04284942E+00 3 25 22 22 # h_1 gamma gamma coupling
1.02186767E+00 3 25 21 21 # h_1 g g coupling
0.00000000E+00 4 25 21 23 # h_1 g g Z coupling
0.00000000E+00 3 25 25 23 # h_1 h_1 Z coupling
Block EFFHIGGSCOUPPLINGS # values of loop-induced couplings
25 22 22 0.33598689E-04 # H-Photon-Photon
25 21 21 0.65965686E-04 # H-Gluon-Gluon
25 22 23 0.00000000E+00 # H-Photon-Z (not yet calculated by SPheno)
Block SPhenoLowEnergy # low energy observables
1 -0.00000000E+00 # T-parameter (1-loop BSM)
2 0.00000000E+00 # S-parameter (1-loop BSM)
3 0.00000000E+00 # U-parameter (1-loop BSM)
20 1.99137438E-23 # (g-2)_e
21 2.00436756E-14 # (g-2)_mu
22 9.10708358E-10 # (g-2)_tau
23 0.00000000E+00 # EDM(e)
24 0.00000000E+00 # EDM(mu)
25 0.00000000E+00 # EDM(tau)
39 -3.57242562E-04 # delta(rho)
Block FlavorKitQFV # quark flavor violating observables
200 3.15000000E-04 # BR(B->X_s gamma)
201 1.00000000E+00 # BR(B->X_s gamma)/BR(B->X_s gamma) SM
-:--- SPheno.spc.SSDM 22% L186 (Fundamental)

```



```

Block FlavorKitFV # quark flavor violating observables
200 3.1500000E-04 # BR(B->X s gamma)
201 1.0000000E-00 # BR(B->X s gamma)/BR(B->X s gamma)_SM
300 5.8929158E-04 # BR(D->nu nu)
301 9.8055696E-01 # BR(D->nu nu)/BR(D->nu nu)_SM
400 5.5724535E-03 # BR(Ds->nu nu)
401 9.9883031E-01 # BR(Ds->nu nu)/BR(Ds->nu nu)_SM
402 5.4460003E-02 # BR(Ds->tau nu)
403 9.0883636E-01 # BR(Ds->tau nu)/BR(Ds->tau nu)_SM
500 5.0585803E-07 # BR(B->nu nu)
501 9.9164564E-01 # BR(B->nu nu)/BR(B->nu nu)_SM
502 1.1250575E-04 # BR(B->tau nu)
503 0.0164564E-01 # BR(B->tau nu)/BR(B->tau nu)_SM
600 6.3232954E-01 # BR(K->nu nu)
601 9.9926792E-01 # BR(K->nu nu)/BR(K->nu nu)_SM
602 2.4764773E-05 # R_K = BR(K->nu nu)/(K->nu nu)
603 2.4761475E-05 # R_K* = BR(K->nu nu)_SM/(K->nu nu)_SM
1900 1.0000000E-00 # Delta(M_B)
1901 1.0005703E-00 # Delta(M_B)/Delta(M_B)_SM
1902 4.2553557E-01 # Delta(M_B)
1903 1.00018573E+00 # Delta(M_B)/Delta(M_B)_SM
4000 1.0015202E-15 # BR(B^0 d->e e)
4001 1.0000046E+00 # BR(B^0 d->e e)/BR(B^0 d->e e)_SM
4002 7.1809117E-14 # BR(B^0 s->e e)
4003 1.0000043E+00 # BR(B^0 s->e e)/BR(B^0 s->e e)_SM
4004 4.5340958E-11 # BR(B^0 d->nu nu)
4005 1.0000064E+00 # BR(B^0 d->nu nu)/BR(B^0 d->nu nu)_SM
4006 3.0711016E-00 # BR(B^0 s->nu nu)
4007 1.0000005E+00 # BR(B^0 s->nu nu)/BR(B^0 s->nu nu)_SM
4008 9.4916154E-00 # BR(B^0 d->tau tau)
4009 1.0000035E+00 # BR(B^0 d->tau tau)/BR(B^0 d->tau tau)_SM
4010 6.5132806E-07 # BR(B^0 s->tau tau)
4011 1.0000005E+00 # BR(B^0 s->tau tau)/BR(B^0 s->tau tau)_SM
5000 1.6414144E-06 # BR(B-> s e)
5001 0.0011625E-01 # BR(B-> s e)/BR(B-> s e)_SM
5002 1.5910105E-00 # BR(B-> s nu nu)
5003 9.9141738E-01 # BR(B-> s nu nu)/BR(B-> s nu nu)_SM
6000 1.1090418E-07 # BR(B-> K nu nu)
6001 9.9913677E-01 # BR(B-> K nu nu)/BR(B-> K nu nu)_SM
6002 1.1090418E-07 # BR(B-> K nu nu)
6003 9.9913677E-01 # BR(B-> K nu nu)/BR(B-> K nu nu)_SM
7000 3.0864209E-05 # BR(B-> nu nu)
7001 1.0000000E+00 # BR(B-> nu nu)/BR(B-> nu nu)_SM
7002 8.0737925E-07 # BR(B->0 nu nu)
7003 1.0000000E+00 # BR(B->0 nu nu)/BR(B->0 nu nu)_SM
8000 1.3084584E-10 # BR(K*+ -> pi*+ nu nu)
8001 1.0000000E+00 # BR(K*+ -> pi*+ nu nu)/BR(K*+ -> pi*+ nu nu)_SM
8002 1.6228873E-43 # BR(K_L -> pi^0 nu nu)
8003 1.0000000E+00 # BR(K_L -> pi^0 nu nu)/BR(K_L -> pi^0 nu nu)_SM
8004 0.0000000E+00 # BR(K*0 L -> e mu)
8005 0.0000000E+00 # BR(K*0 L -> e mu)/BR(K*0 L -> e mu)_SM
9100 1.0464305E-15 # Delta(M_K)
9102 9.9999342E-01 # Delta(M_K)/Delta(M_K)_SM
9103 1.6437074E-03 # epsilon_L K
9104 1.0000000E+00 # epsilon_L K/epsilon_L K_SM

Block FlavorKitLV # lepton flavor violating observables
701 0.0000000E+00 # BR(mu-> gamma)
702 0.0000000E+00 # BR(tau-> gamma)
703 0.0000000E+00 # BR(tau->nu gamma)
800 0.0000000E+00 # CR(mu-> A)
801 0.0000000E+00 # CR(mu-> T)
802 0.0000000E+00 # CR(mu-> S)
803 0.0000000E+00 # CR(mu-> Sb)
804 0.0000000E+00 # CR(mu-> Au)
805 0.0000000E+00 # CR(mu-> Pb)
901 0.0000000E+00 # CR(mu->3e)
902 0.0000000E+00 # BR(tau->3e)
903 0.0000000E+00 # BR(tau->3mu)
904 0.0000000E+00 # BR(tau -> e- mu- nu-)
905 0.0000000E+00 # BR(tau -> mu- e- e-)
906 0.0000000E+00 # BR(tau -> nu- mu- nu-)
907 0.0000000E+00 # BR(tau -> mu- e- e-)
1001 0.0000000E+00 # BR(Z-> nu nu)
1002 0.0000000E+00 # BR(Z->e tau)
1003 0.0000000E+00 # BR(Z->mu tau)
1101 0.0000000E+00 # BR(h->nu nu)
1102 0.0000000E+00 # BR(h->tau tau)
1103 0.0000000E+00 # BR(h->mu tau)
2001 0.0000000E+00 # BR(tau->nu pi)
2002 0.0000000E+00 # BR(tau->nu eta)
2003 0.0000000E+00 # BR(tau->nu eta')
2004 0.0000000E+00 # BR(tau->nu pi)
2005 0.0000000E+00 # BR(tau->nu eta)
2006 0.0000000E+00 # BR(tau->nu eta')

Block FACIEF Q= 1.6000000E+02 # Wilson coefficients at scale 0
0300 4422 00 0 -0.1646170E-00 # coeffCTm
0300 4422 00 2 -0.1646170E-00 # coeffCT
0300 4322 00 2 -0.3737089E-10 # coeffCTp
0300 4422 00 1 0.0000000E+00 # coeffCTNP
0300 4322 00 1 -0.3737089E-10 # coeffCTpNP
0300 6421 00 0 -0.8247673E-09 # coeffCBm
0300 6421 00 2 -0.8247673E-09 # coeffCB
0300 6321 00 2 -0.1930948E-10 # coeffCTp
0300 6421 00 1 0.0000000E+00 # coeffCBNP
0300 6321 00 1 -0.1930948E-10 # coeffCBNP
03051111 4133 00 0 0.1010354E-00 # coeffCbmSM
03051111 4133 00 2 0.1010354E-00 # coeffCbm
03051111 4213 00 0 0.2161505E-14 # coeffCbmW
03051111 4135 00 1 0.0000000E+00 # coeffCbmW

new bar options new.pdf Font

```



```
Block FlavorKitQV # quark flavor violating observables
200 3.1500000E-04 # BR(B->X gamma)
201 1.0000000E+00 # BR(B->X gamma)/BR(B->X gamma)_SM
300 5.8929158E-04 # BR(D->nu nu)
301 9.8925696E-01 # BR(D->nu nu)/BR(D->nu nu)_SM
400 5.5724535E-03 # BR(Ds->nu nu)
401 9.9883031E-01 # BR(Ds->nu nu)/BR(Ds->nu nu)_SM
402 5.4460003E-02 # BR(Ds->tau nu)
403 9.0883636E-01 # BR(Ds->tau nu)/BR(Ds->tau nu)_SM
500 5.0285803E-07 # BR(B->nu nu)
501 9.9164564E-01 # BR(B->nu nu)/BR(B->nu nu)_SM
502 1.1250575E-04 # BR(B->tau nu)
503 0.0164564E-01 # BR(B->tau nu)/BR(B->tau nu)_SM
600 6.3232954E-01 # BR(K->nu nu)
601 9.9925793E-01 # BR(K->nu nu)/BR(K->nu nu)_SM
602 2.4764773E-05 # R_K = BR(K->nu nu)/(K->nu nu)
603 2.4761475E-05 # R_K^SM = BR(K->nu nu)_SM/(K->nu nu)_SM
1900 1.0000307E+00 # Delta(M_B)
1901 1.0005703E+00 # Delta(M_B)/Delta(M_B)_SM
1902 2.2553575E-01 # Delta(M_B)
1903 1.0001573E+00 # Delta(M_B)/Delta(M_B)_SM
4000 1.0015202E-15 # BR(B^0_d->e e)
4001 1.0000046E+00 # BR(B^0_d->e e)/BR(B^0_d->e e)_SM
4002 7.1809117E-14 # BR(B^0_s->e e)
4003 1.0000043E+00 # BR(B^0_s->e e)/BR(B^0_s->e e)_SM
4004 4.5340958E-11 # BR(B^0_d->nu mu)
4005 1.0000064E+00 # BR(B^0_d->nu mu)/BR(B^0_d->nu mu)_SM
4006 3.0711016E-09 # BR(B^0_s->nu mu)
4007 1.0000005E+00 # BR(B^0_d->tau tau)/BR(B^0_d->tau tau)_SM
4008 9.4916154E-09 # BR(B^0_s->tau tau)
4009 1.0000035E+00 # BR(B^0_d->tau tau)/BR(B^0_d->tau tau)_SM
4010 6.5132806E-07 # BR(B^0_s->tau tau)
4011 1.0000005E+00 # BR(B^0_s->tau tau)/BR(B^0_s->tau tau)_SM
5000 1.6414144E-06 # BR(B->S e e)
5001 0.0017425E-01 # BR(B->S e e)/BR(B->S e e)_SM
5002 1.5910105E-06 # BR(B->S nu mu)
5003 9.9141738E-01 # BR(B->S nu mu)/BR(B->S nu mu)_SM
6000 1.1090418E-07 # BR(B->K mu mu)
6001 9.9913077E-01 # BR(B->K mu mu)/BR(B->K mu mu)_SM
6002 1.1090418E-07 # BR(B->K mu mu)
6003 9.9913077E-01 # BR(B->K mu mu)/BR(B->K mu mu)_SM
7000 3.0862096E-05 # BR(B->nu nu)
7001 1.0000000E+00 # BR(B->nu nu)/BR(B->nu nu)_SM
7002 8.0737925E-07 # BR(B->0 nu nu)
7003 1.0000000E+00 # BR(B->0 nu nu)/BR(B->0 nu nu)_SM
8000 1.3084584E-10 # BR(K^+->pi^+ nu mu)
8001 1.0000000E+00 # BR(K^+->pi^+ nu mu)/BR(K^+->pi^+ nu mu)_SM
8002 1.6228873E-43 # BR(K_L->pi^0 nu mu)
8003 1.0000000E+00 # BR(K_L->pi^0 nu mu)/BR(K_L->pi^0 nu mu)_SM
8004 0.0000000E+00 # BR(K^0_L->e mu)
8005 0.0000000E+00 # BR(K^0_L->e mu)/BR(K^0_L->e mu)_SM
1000 1.0461093E-15 # Delta(M_K)/Delta(M_K)_SM
9102 9.9999342E-01 # Delta(M_K)
9103 1.6437074E-03 # epsilon_K
10000 0.0000000E+00 # CKM CKM^CP
Block FlavorKitLV # lepton flavor violating observables
701 0.0000000E+00 # BR(tau->e gamma)
702 0.0000000E+00 # BR(tau->mu gamma)
703 0.0000000E+00 # BR(tau->tau gamma)
800 0.0000000E+00 # CR(mu->A)
801 0.0000000E+00 # CR(mu->Ti)
802 0.0000000E+00 # CR(mu->Sr)
803 0.0000000E+00 # CR(mu->Sb)
804 0.0000000E+00 # CR(mu->Au)
805 0.0000000E+00 # CR(mu->Pb)
901 0.0000000E+00 # BR(mu->3e)
902 0.0000000E+00 # BR(tau->3e)
903 0.0000000E+00 # BR(tau->3mu)
904 0.0000000E+00 # BR(tau->e mu mu)
905 0.0000000E+00 # BR(tau->mu mu e)
906 0.0000000E+00 # BR(tau->nu mu mu)
907 0.0000000E+00 # BR(tau->nu mu e)
1001 0.0000000E+00 # BR(Z->nu nu)
1002 0.0000000E+00 # BR(Z->e tau)
1003 0.0000000E+00 # BR(Z->mu tau)
1004 0.0000000E+00 # BR(h->tau tau)
1005 0.0000000E+00 # BR(h->mu tau)
2001 0.0000000E+00 # BR(tau->nu pi)
2002 0.0000000E+00 # BR(tau->nu eta)
2003 0.0000000E+00 # BR(tau->nu eta')
2004 0.0000000E+00 # BR(tau->nu pi)
2005 0.0000000E+00 # BR(tau->nu eta)
2006 0.0000000E+00 # BR(tau->nu eta')
```

```
Block FACIEP Q= 1.0000000E+02 # Wilson coefficients at scale 0
```

```
0305 4422 00 0 -0.1646170E-06 # coeffCTm
0305 4422 00 2 -0.1646170E-06 # coeffCT
0305 4322 00 2 -0.3737089E-10 # coeffCTp
0305 4422 00 1 0.0000000E+00 # coeffCTNP
0305 4322 00 1 -0.3737089E-10 # coeffCTpNP
0305 6421 00 0 -0.8247673E-09 # coeffCBm
0305 6421 00 2 -0.8247673E-09 # coeffCB
0305 6321 00 2 -0.1939948E-10 # coeffCBp
0305 6421 00 1 0.0000000E+00 # coeffCBNP
0305 6321 00 1 -0.1939948E-10 # coeffCBpNP
03051111 4133 00 0 -0.1010354E-06 # coeffCDeSM
03051111 4133 00 2 0.1010354E-06 # coeffCDe
03051111 4233 00 0 -0.2161505E-14 # coeffCDeW
03051111 4135 00 1 0.0000000E+00 # coeffCDeW
```

```
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```

Block FlavorKitLV # lepton flavor violating observables

```
701 0.0000000000E+00 # BR(mu->e gamma)
702 0.0000000000E+00 # BR(tau->e gamma)
703 0.0000000000E+00 # BR(tau->mu gamma)
```

```

01050105 3232 00 0 0.00000000E+00 # coeffBB_SRRSM
01050105 3132 00 0 0.00000000E+00 # coeffBB_SLRSM
01050105 4141 00 0 0.00000000E+00 # coeffBB_VLLSM
01050105 4242 00 0 0.00000000E+00 # coeffBB_VRRSM
01050105 4142 00 0 0.00000000E+00 # coeffBB_VLRSM
01050105 4343 00 0 0.00000000E+00 # coeffBB_TLLSM
01050105 4444 00 0 0.00000000E+00 # coeffBB_TRRSM
03050305 3131 00 0 0.00000000E+00 # coeffBsBs_SLLSM
03050305 3232 00 0 0.00000000E+00 # coeffBsBs_SRRSM
03050305 3132 00 0 0.00000000E+00 # coeffBsBs_SLRSM
03050305 4141 00 0 0.00000000E+00 # coeffBsBs_VLLSM
03050305 4242 00 0 0.00000000E+00 # coeffBsBs_VRRSM
03050305 4142 00 0 0.00000000E+00 # coeffBsBs_VLRSM
03050305 4343 00 0 0.00000000E+00 # coeffBsBs_TLLSM
03050305 4444 00 0 0.00000000E+00 # coeffBsBs_TRRSM

```

Block TREELEVELUNITARITY

```

0 1.00000000E+00 # Tree-level unitarity limits fulfilled or not
1 1.67207372E-02 # Maximal scattering eigenvalue

```

Block TREELEVELUNITARITYwTRILINEARS

```

0 1.00000000E+00 # Tree-level unitarity limits fulfilled or not
1 1.61576897E-02 # Maximal scattering eigenvalue
2 2.00000000E+03 # best scattering energy
11 1.00000000E+03 # min scattering energy
12 2.00000000E+03 # max scattering energy
13 5.00000000E+00 # steps

```

DECAY 4 3.82261015E-13 # Fu_2

#	BR	NDA	ID1	ID2	ID3	
#	BR	NDA	ID1	ID2		
	3.05502575E-02	3	2	-1	1	# BR(Fu_2 -> Fu_1 Fd_1^* Fd_1)
	5.45954987E-01	3	2	-1	3	# BR(Fu_2 -> Fu_1 Fd_1^* Fd_2)
	1.56486313E-03	3	2	-3	1	# BR(Fu_2 -> Fu_1 Fd_2^* Fd_1)
	2.79270154E-02	3	2	-3	3	# BR(Fu_2 -> Fu_1 Fd_2^* Fd_2)
	1.07295183E-02	3	1	-11	12	# BR(Fu_2 -> Fd_1 Fe_1^* Fv_1)
	1.01645236E-02	3	1	-13	14	# BR(Fu_2 -> Fd_1 Fe_2^* Fv_2)
	1.91744771E-01	3	3	-11	12	# BR(Fu_2 -> Fd_2 Fe_1^* Fv_1)
	1.81364064E-01	3	3	-13	14	# BR(Fu_2 -> Fd_2 Fe_2^* Fv_2)

DECAY 6 1.55526925E+00 # Fu_3

#	BR	NDA	ID1	ID2	
	1.67597777E-03	2	3	24	# BR(Fu_3 -> Fd_2 Vwp)
	9.98288583E-01	2	5	24	# BR(Fu_3 -> Fd_3 Vwp)

```

2.15252618E-03 3 2 -12 11 # BR(Fd_3 -> Fu_1 Fv_1^* Fe_1 )
2.14490780E-03 3 2 -14 13 # BR(Fd_3 -> Fu_1 Fv_2^* Fe_2 )
5.83957751E-04 3 2 -16 15 # BR(Fd_3 -> Fu_1 Fv_3^* Fe_3 )
1.59069889E-01 3 4 -12 11 # BR(Fd_3 -> Fu_2 Fv_1^* Fe_1 )
1.58104211E-01 3 4 -14 13 # BR(Fd_3 -> Fu_2 Fv_2^* Fe_2 )
1.96896118E-02 3 4 -16 15 # BR(Fd_3 -> Fu_2 Fv_3^* Fe_3 )

```

```

DECAY 25 6.42252863E-03 # hh
# BR NDA ID1 ID2
1.93338706E-03 2 22 22 # BR(hh -> VP VP )
5.96211424E-02 2 21 21 # BR(hh -> VG VG )
2.82490956E-02 2 23 23 # BR(hh -> VZ VZ )
2.33261695E-01 2 -24 24 # BR(hh -> Vwp^* Vwp_virt )
1.33790716E-04 2 -3 3 # BR(hh -> Fd_2^* Fd_2 )
3.58433068E-01 2 -5 5 # BR(hh -> Fd_3^* Fd_3 )
1.45065926E-04 2 -13 13 # BR(hh -> Fe_2^* Fe_2 )
4.18774507E-02 2 -15 15 # BR(hh -> Fe_3^* Fe_3 )
1.68995783E-02 2 -4 4 # BR(hh -> Fu_2^* Fu_2 )
2.59445280E-01 2 6666635 6666635 # BR(hh -> ss ss )

```

```

DECAY1L 4 1.11448989E-23 # Fu_2
# BR NDA ID1 ID2
9.80578882E-01 2 21 21 # BR(Fu_2 -> Fu_1 VG )
1.94211179E-02 2 22 22 # BR(Fu_2 -> Fu_1 VP )

```

```

DECAY1L 6 1.40218346E+00 # Fu_3
# BR NDA ID1 ID2
1.67434891E-03 2 24 24 # BR(Fu_3 -> Fd_2 Vwp )
9.98290247E-01 2 24 24 # BR(Fu_3 -> Fd_3 Vwp )

```

```

DECAY1L 3 1.38160366E-20 # Fd_2
# BR NDA ID1 ID2
9.93677595E-01 2 21 21 # BR(Fd_2 -> Fd_1 VG )
6.32240539E-03 2 22 22 # BR(Fd_2 -> Fd_1 VP )

```

```

DECAY1L 5 4.50364203E-14 # Fd_3
# BR NDA ID1 ID2
2.05693613E-02 2 21 21 # BR(Fd_3 -> Fd_1 VG )
9.74708472E-01 2 21 21 # BR(Fd_3 -> Fd_2 VG )
4.62332156E-03 2 22 22 # BR(Fd_3 -> Fd_2 VP )

```

```

DECAY1L 25 8.26580363E-03 # hh
# BR NDA ID1 ID2
3.55304437E-04 2 3 3 # BR(hh -> Fd_2^* Fd_2 )
6.75629528E-01 2 -5 5 # BR(hh -> Fd_3^* Fd_3 )
1.19436199E-04 2 -13 13 # BR(hh -> Fe_2^* Fe_2 )

```

→ $\text{BR}(h \rightarrow SS) = 26\%$

Run micrOMEGAs

```
micromegas/SSDM/CalcOmega_with_DDetection_M0v5 SPheno.spc.SSDM
```

```
Masses of odd sector Particles:
```

```
-Ss : MSs = 28.4 ||
```

```
PROCESS: -Ss,-Ss->AllEven,1*x{h,g,A,Z,Wp,Wm,nu1,Nu1,nu2,Nu2,nu3,Nu3,d1,D1,d2,D2,d3,D3,u1,U1,u2,U2,u3,U3,e1,E1,e2,E2,e3,  
E3
```

```
Xf=1.64e+01 Omega h^2=2.28e+01
```

```
# Channels which contribute to 1/(omega) more than 1%.
```

```
# Relative contributions in % are displayed
```

```
85% -Ss -Ss ->d3 D3
```

```
8% -Ss -Ss ->e3 E3
```

```
4% -Ss -Ss ->u2 U2
```

```
2% -Ss -Ss ->g g
```

```
==== Calculation of CDM-nucleons amplitudes ====
```

```
TREE LEVEL
```

```
PROCESS: QUARKS,-Ss->QUARKS,-Ss{d1,D1,d2,D2,d3,D3,u1,U1,u2,U2,u3,U3
```

```
Delete diagrams with _S0_!=1,_V5_,A
```

```
....
```

```
CDM-nucleon cross sections[pb]:
```

```
proton SI 2.407E-09 SD 0.000E+00
```

```
neutron SI 2.471E-09 SD 0.000E+00
```

```
===== Direct Detection =====
```

```
73Ge: Total number of events=1.29E-03 /day/kg
```

```
Number of events in 10 - 50 KeV region=4.19E-04 /day/kg
```

```
131Xe: Total number of events=2.66E-03 /day/kg
```

Automatic generation of scotogenic models

Python program: minimal-lagrangians

Simon May, <https://arxiv.org/pdf/2003.08037> [CPC]

```
pip install minimal-lagrangians
```

- As the program was originally written for the study of minimal darkmatter models with radiative neutrino masses: D. Restrepo, O. Zapata, C. E. Yaguna, Models with radiative neutrino masses and viable dark matter candidates, arXiv:1308.3655 [JHEP]
- Lagrangians for the individual models.
- Model files for SARAH can be constructed automatically from the specified field content, which can be tedious if done manually

Thus, minimal-lagrangians enables rapid phenomenological studies using SARAH and, successively, further tools like SPheno, and micrOMEGAs → Sec. 8: Outlook

```
BSMModel('T1-3-B', (  
  FermionField("Ψ", 1, Y= 0),  
  FermionField("Ψ'", 2, Y= 1),  
  ScalarField ("phi", 3, Y= 0),  
  FermionField("Ψ'", 2, Y=-1),6), (0, 2)), # = -2 is equivalent to= 2
```

Computer tools in particle physics

Information

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- [CINVESTAV, México City \(México\) 2015](#)
- [IFIC, Valencia \(Spain\) 2016](#)
- [Universidad de Antioquia, Medellín \(Colombia\) 2016](#)
- [IFIC, Valencia \(Spain\) 2017](#)

References

The course focuses on the material contained in the following notes:

[Computer tools in particle physics, A. Vicente, arXiv:1507.06349 \[PDF\]](#)

For two-loops RGEs see also:

["Exploring new models in all detail with SARAH", Florian Staub, arXiv:1503.04200 \[PDF\]](#)

SARAH:

["SARAH 4: A tool for \(not only SUSY\) model builders", Florian Staub, arXiv:1309.7223 \[PDF\]](#)

About

This is the website for the course 'Computer tools in particle physics'.

Links

V1.0 August 2009: Susy Only
V4.0 September 2013: non-Susy
V4.14.2 (Transferred to W.Porod)

- [SARAH](#)
- [SPheno](#)
- [MicrOMEGAs](#)
- [MadGraph](#)
- [MadAnalysis](#)
- [FlavorKit](#)

Contact

Avelino Vicente
IFIC (CSIC/U. Valencia)
Office B-6-0

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✱ Computer tools in particle physics

Information

This is the website for the course [Computer tools in particle physics](#) by Avelino Vicente, to take place at [Instituto de Física Corpuscular](#) (CSIC/Universidad de Valencia).

Dates: Monday 22/05/2017 - Friday 26/05/2017

Place: IFIC - Sala de Audiovisuales (Nave experimental)

Time: 15:00

Duration: 1.5 h for the first session and 1 h for the rest

Material and required programs

This will be a hands-on course, where all participants are encouraged to run all codes in their own laptops. The only required programas are [Mathematica](#), a [LaTeX compiler](#) and [Fortran 90 and C++ compilers](#). If you wish to fully participate please download the following files:

- For lecture 1: [run_sarah_Scotogenic.nb](#) and [Scotogenic.tar.gz](#)
- For lecture 2: [micromegas_4.2.5.tgz](#)
- For lecture 4: [run_sarah_DarkBS.nb](#), [DarkBS.tar.gz](#) and [plotDarkBS.txt](#)

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About

This is the website for the course [Computer tools in particle physics](#). IFIC (CSIC/U. Valencia), May 22nd - 26th, 2017.

Links

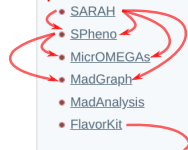
- [SARAH](#)
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- [FlavorKit](#)

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Input/Output Code



Observables already in FlavorKit

Lepton flavor	Quark flavor
$\ell_\alpha \rightarrow \ell_\beta \gamma$	$B_{s,d}^0 \rightarrow \ell^+ \ell^-$
$\ell_\alpha \rightarrow 3 \ell_\beta$	$\bar{B} \rightarrow X_s \gamma$
$\mu - e$ conversion in nuclei	$\bar{B} \rightarrow X_s \ell^+ \ell^-$
$\tau \rightarrow P \ell$	$\bar{B} \rightarrow X_{d,s} \nu \bar{\nu}$
$h \rightarrow \ell_\alpha \ell_\beta$	$B \rightarrow K \ell^+ \ell^-$
$Z \rightarrow \ell_\alpha \ell_\beta$	$K \rightarrow \pi \nu \bar{\nu}$
	$\Delta M_{B_{s,d}}$
	ΔM_K and ε_K
	$P \rightarrow \ell \nu$

Ready to be computed in your favourite model!

Observables already in FlavorKit

Lepton flavor	Quark flavor
$\ell_\alpha \rightarrow \ell_\beta \gamma$	$B_{s,d}^0 \rightarrow \ell^+ \ell^-$
$\ell_\alpha \rightarrow 3 \ell_\beta$	$\bar{B} \rightarrow X_s \gamma$
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$h \rightarrow \ell_\alpha \ell_\beta$	$B \rightarrow K \ell^+ \ell^-$
$Z \rightarrow \ell_\alpha \ell_\beta$	$K \rightarrow \pi \nu \bar{\nu}$
	$\Delta M_{B_{s,d}}$
	ΔM_K and ε_K
	$P \rightarrow \ell \nu$

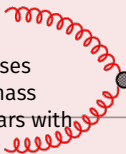
Also in SARAH

S, T, U

One-loop corrections to All masses

Two-loop corrections to Higgs mass

Gluon fusion production of scalars with
proper output for MadGraph



Ready to be computed in your favourite model!

Computer tools in particle physics

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Contact

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IFIC (CSIC/U. Valencia)
Office B-6-0

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Models already in SARAH

Supersymmetric Models

- MSSM [in several versions]
- NMSSM [in several versions]
- Near-to-minimal SSM (near-MSSM)
- General singlet extended SSM (SMSSM)
- DiracNMSSM
- Triplet extended MSSM/NMSSM
- Several models with R-parity violation
- Several U(1)-extended models
- Secluded MSSM
- Several B-L extended models
- Inverse and linear seesaws
- MSSM/NMSSM with Dirac Gauginos
- Minimal R-Symmetric SSM
- Minimal Dirac Gaugino SSM
- Seesaws I-II-III [SU(5) versions]
- Left-right symmetric model
- Quiver model
- Models with vector-like superfields

Non-Supersymmetric Models

- Standard Model
- Two Higgs doublet models (including inert)
- Singlet extensions
- Triplet extensions
- U(1) extensions
- SM extended by a scalar color octet
- Gauged Two Higgs doublet model
- Singlet extended SM
- Singlet Scalar DM
- Singlet-Doublet DM
- Models with vector-like fermions
- Model with a scalar SU(2) 7-plet
- Leptoquark models
- Left-right models
- 331 models (with and without exotics)
- Georgi-Machacek model

More info: <http://sarah.hepforge.org/>

Models already in SARAH

Supersymmetric Models

Always check any version of SARAH and SPheno with this one!

- Minimal Supersymmetric Standard Model (MSSM)
- NMSSM (in several versions)
- Near-minimal SSM (near-MSSM)
- General singlet extended SSM (SMSSM)
- DiracNMSSM
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Non-Supersymmetric Models

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- Left-right symmetric model
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Non-Supersymmetric Models

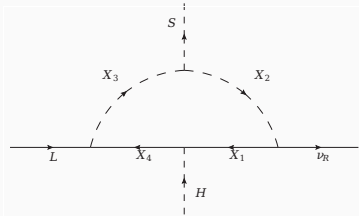
- Standard Model
- Two Higgs doublet models (including inert)
- Singlet extensions
- Triplet extensions
- U(1) extensions
- SM extended by a scalar color octet
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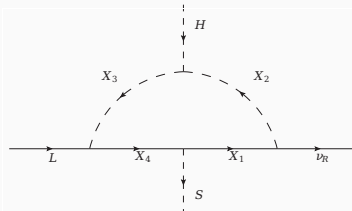


Dirac neutrino masses

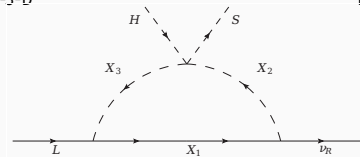
One loop topologies $U(1)_{B-L} \oplus Z_2 \oplus Z_2$



T1-3-D

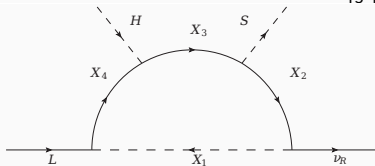


T1-3-E

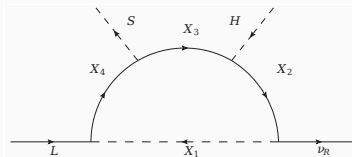


T3-1-A

Chang-Yuan Yao and Gui-Jun Ding, arXiv:1802.05231 [PRD]

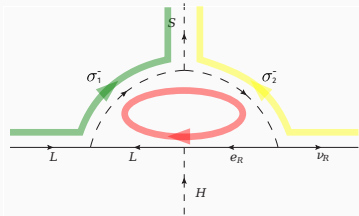


T1-2-A

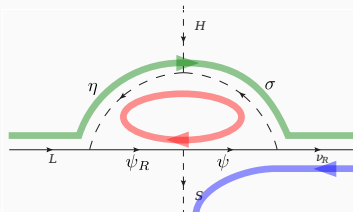


T1-2-B

One loop topologies $U(1)_{B-L}$ only!



T1-3-D

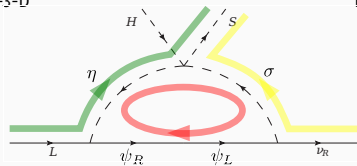


T1-3-E

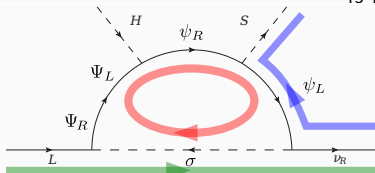
$\psi_{L,R} \rightarrow$ Singlet fermions (vector-like)

$\sigma \rightarrow$ Singlet scalar

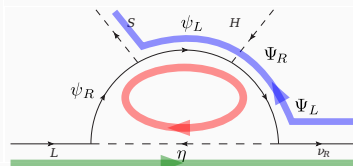
with J. Calle, C. Yaguna, and O. Zapata, arXiv:1812.05523 [PRD]



T3-1-A

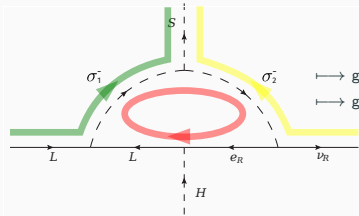


T1-2-A



T1-2-B

One loop topologies $U(1)_{B-L}$ only! with J. Calle, C. Yaguna, and O. Zapata, arXiv:1812.05523 [PRD]



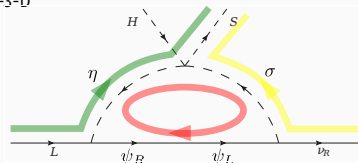
T1-3-D

→ generalization to two and three loops: S. Saad arXiv:1902.07259 [NPB]
 → generalization to $U(1)_R$: *et al*, S. Saad arXiv:1904.07407

Fields: f_i	L, e_R	ν_{R3}	ν_{R2}	ν_{R1}	S	H
$U(1)_{B-L}$	-1	-4	-4	+5	+3	0

$\psi_{L,R} \rightarrow$ Singlet fermions (vector-like)

$\sigma \rightarrow$ Singlet scalar



T3-1-A

Anomaly cancellation conditions

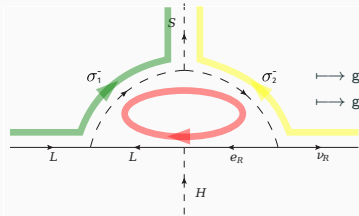
$$\sum_i \nu_{Ri} = -3$$

$$\sum_i \nu_{Ri}^3 = -3$$

Three-level cancellation conditions

$$\cancel{(\nu_R)^\dagger L \cdot H}$$

One loop topologies $U(1)_{B-L}$ only! with J. Calle, C. Yaguna, and O. Zapata, arXiv:1812.05523 [PRD]



T1-3-D

→ generalization to two and three loops: S. Saad arXiv:1902.07259 [NPB]

→ generalization to $U(1)_R$: *et al*, S. Saad arXiv:1904.07407

Fields: f_i	L, e_R	ν_{R3}	ν_{R2}	ν_{R1}	σ_1^-	σ_2^-	S	H
$U(1)_{B-L}$	-1	-4	-4	+5	-2	-5	+3	0

Anomaly cancellation conditions

$\sigma \rightarrow$ Singlet scalar

$$\sum_i \nu_{Ri} = -3$$

$$\sum_i \nu_{Ri}^3 = -3$$

Three-level cancellation conditions

$$\frac{\nu_R \nu_R}{(\nu_R)^\dagger L \cdot H}$$

```
cp -r BSM/SARAH/Models/B-L/ SARAH/Models/  
math << EOF  
<<./SARAH/SARAH.m  
Start["B-L/DZ"]  
MakeSPheno []  
EOF
```

```
cp -r SARAH/Output/B-L-DZ/EWSB/SPheno SPheno/BLDZ  
cd SPheno  
make Model=BLDZ  
cd .. # Return to parent directory
```

```
cp BSM/Input_Files/LesHouches.in.BLDZ .
```

LesHouches.in.BLDZ: Fix options!

```
55 1      # Calculate loop corrected masses  
50 0      # Majorana phases: use only positive masses  
520 0     # Write effective Higgs couplings
```

```
cp -r BSM/SARAH/Models/B-L/ SARAH/Models/  
math << EOF  
<<./SARAH/SARAH.m  
Start["B-L/DZ"]  
MakeSPheno []  
EOF
```

```
cp -r SARAH/Output/B-L-DZ/EWSB/SPheno SPheno/BLDZ  
cd SPheno  
make Model=BLDZ  
cd .. # Return to parent directory
```

```
cp BSM/Input_Files/LesHouches.in.BLDZ .
```

```
SPheno/bin/SPhenoBLDZ LesHouches.in.BLDZ
```

cat SPheno.spc.BLDZ

```
Block MASS # Mass spectrum
# PDG code mass particle
      25      1.24861947E+02 # hh_1
      35      1.71464282E+03 # hh_2
  900037      2.00000000E+03 # Hm_2
  900038      3.00000000E+03 # Hm_3
      22      0.00000000E+00 # VP
      23      9.11887000E+01 # VZ
      21      0.00000000E+00 # VG
      24      7.96796394E+01 # VWm
      31      2.57196423E+03 # VZp
       1      5.00000000E-03 # Fd_1
      .
      .
      .
      15      1.77669000E+00 # Fe_3
      12      0.00000000E+00 # Fv_1
      14     -1.61994502E-31 # Fv_2
      16      4.42048291E-14 # Fv_3
  8810012     -4.42048291E-14 # Fv_4
  8810014      2.08300541E-10 # Fv_5
  8810016     -2.08300541E-10 # Fv_6
```

EXTRA

2nd Chuck Norris fact of the day

*Chuck Norris can run collider
simulations with MadGraph on
an abacus*

From A. Vicente

Backup slides

Preliminars

Computer tools in particle physics

Information

This is the website for the course 'Computer tools in particle physics' by [Avelino](#) Vicente.

- [CINVESTAV, México City \(México\) 2015](#)
- [IFIC, Valencia \(Spain\) 2016](#)
- [Universidad de Antioquia, Medellín \(Colombia\) 2016](#)
- [IFIC, Valencia \(Spain\) 2017](#)

References

The course focuses on the material contained in the following notes:

[Computer tools in particle physics, A. Vicente, arXiv:1507.06349 \[PDF\]](#)

For two-loops RGEs see also:

["Exploring new models in all detail with SARAH", Florian Staub, arXiv:1503.04200 \[PDF\]](#)

SARAH:

["SARAH 4: A tool for \(not only SUSY\) model builders", Florian Staub, arXiv:1309.7223 \[PDF\]](#)

About

This is the website for the course 'Computer tools in particle physics'.

Links

V1.0 August 2009: Susy Only
V4.0 September 2013: non-Susy
V4.14.2 (Transferred to W.Porod)

- [SARAH](#)
- [SPheno](#)
- [MicrOMEGAs](#)
- [MadGraph](#)
- [MadAnalysis](#)
- [FlavorKit](#)

Contact

Avelino Vicente
IFIC (CSIC/U. Valencia)
Office B-6-0

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✱ Computer tools in particle physics

Information

This is the website for the course [Computer tools in particle physics](#) by Avelino Vicente, to take place at [Instituto de Física Corpuscular](#) (CSIC/Universidad de Valencia).

Dates: Monday 22/05/2017 - Friday 26/05/2017

Place: IFIC - Sala de Audiovisuales (Nave experimental)

Time: 15:00

Duration: 1.5 h for the first session and 1 h for the rest

Material and required programs

This will be a hands-on course, where all participants are encouraged to run all codes in their own laptops. The only required programas are [Mathematica](#), a [LaTeX compiler](#) and [Fortran 90 and C++ compilers](#). If you wish to fully participate please download the following files:

- For lecture 1: [run_sarah_Scotogenic.nb](#) and [Scotogenic.tar.gz](#)
- For lecture 2: [micromegas_4.2.5.tgz](#)
- For lecture 4: [run_sarah_DarkBS.nb](#), [DarkBS.tar.gz](#) and [plotDarkBS.txt](#)

You should also download the latest versions of the codes we are going to use (exception: for lecture 2 we will use an old version of MicrOMEGAS, see above). You can find them in their official websites (links on your right). Finally, the slides of the course are available here: [introduction](#), [lecture 1](#), [lecture 2](#), [lecture 3](#), [lecture 4](#) and [lecture 5](#).

References

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[Computer tools in particle physics](#), A. Vicente, [arXiv:1507.06349](#)

About

This is the website for the course [Computer tools in particle physics](#). IFIC (CSIC/U. Valencia), May 22nd - 26th, 2017.

Links

- [SARAH](#)
- [SPheno](#)
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- [FlavorKit](#)

Contact

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Input/Output Code

Observables already in FlavorKit

Lepton flavor	Quark flavor
$\ell_\alpha \rightarrow \ell_\beta \gamma$	$B_{s,d}^0 \rightarrow \ell^+ \ell^-$
$\ell_\alpha \rightarrow 3 \ell_\beta$	$\bar{B} \rightarrow X_s \gamma$
$\mu - e$ conversion in nuclei	$\bar{B} \rightarrow X_s \ell^+ \ell^-$
$\tau \rightarrow P \ell$	$\bar{B} \rightarrow X_{d,s} \nu \bar{\nu}$
$h \rightarrow \ell_\alpha \ell_\beta$	$B \rightarrow K \ell^+ \ell^-$
$Z \rightarrow \ell_\alpha \ell_\beta$	$K \rightarrow \pi \nu \bar{\nu}$
	$\Delta M_{B_{s,d}}$
	ΔM_K and ε_K
	$P \rightarrow \ell \nu$

Ready to be computed in your favourite model!

Observables already in FlavorKit

Lepton flavor	Quark flavor
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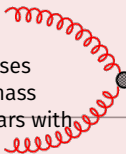
Also in SARAH

S, T, U

One-loop corrections to All masses

Two-loop corrections to Higgs mass

Gluon fusion production of scalars with
proper output for MadGraph



Ready to be computed in your favourite model!

Computer tools in particle physics

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IFIC (CSIC/U. Valencia)
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Models already in SARAH

Supersymmetric Models

- MSSM [in several versions]
- NMSSM [in several versions]
- Near-to-minimal SSM (near-MSSM)
- General singlet extended SSM (SMSSM)
- DiracNMSSM
- Triplet extended MSSM/NMSSM
- Several models with R-parity violation
- Several U(1)-extended models
- Secluded MSSM
- Several B-L extended models
- Inverse and linear seesaws
- MSSM/NMSSM with Dirac Gauginos
- Minimal R-Symmetric SSM
- Minimal Dirac Gaugino SSM
- Seesaws I-II-III [SU(5) versions]
- Left-right symmetric model
- Quiver model
- Models with vector-like superfields

Non-Supersymmetric Models

- Standard Model
- Two Higgs doublet models (including inert)
- Singlet extensions
- Triplet extensions
- U(1) extensions
- SM extended by a scalar color octet
- Gauged Two Higgs doublet model
- Singlet extended SM
- Singlet Scalar DM
- Singlet-Doublet DM
- Models with vector-like fermions
- Model with a scalar SU(2) 7-plet
- Leptoquark models
- Left-right models
- 331 models (with and without exotics)
- Georgi-Machacek model

More info: <http://sarah.hepforge.org/>

Models already in SARAH

Supersymmetric Models

Always check any version of SARAH and SPheno with this one!

- Minimal Supersymmetric Standard Model (MSSM)
- NMSSM (in several versions)
- Near-minimal SSM (near-MSSM)
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